

Type-Aware Failure Prediction for Chart Reasoning in VLMs

Mid-term status, for TA guidance

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The question

Two ways to answer a chart question wrongly

Structural failure (misread)

- Wrong bar, misjudged tick, miscounted slices
- The answer *is* in the image
- Repair: look again (crop, re-ask)

Fabrication

- States a value or label the chart does not contain
- Looking again cannot help
- Repair: abstain

Today's reliability tools collapse both into one "do not trust" score.

Type Separability Hypothesis: the model's internal state *before* it answers is linearly different for a coming misread and a coming fabrication. A collapse of the two into one signal is a reportable outcome, not a failure.

Experiments at a glance

	What it asks	Status
E1	Can a probe predict <i>whether</i> the answer will be wrong?	done
E2	Is the probe just reading surface features of the input?	partial
E3	Are the misread and fabrication signals different (cross-transfer)?	done (synthetic)
E4	Does the failure type decide which repair works?	code ready
E5	A controller that picks the repair from the probes	Oct

Two data arms: **ChartQA** (real charts, naturally occurring errors) and a **synthetic bar-chart arm** (errors induced on purpose).

Pipeline

Setup

Models (frozen, fp16, greedy)

- Qwen2.5-VL-7B (28 layers, $d = 3584$)
- LLaVA-NeXT-7B, Mistral LM (32 layers, $d = 4096$)
- ChartGemma dropped

Data

- ChartQA test: 2,500 questions (1,250 human, 1,250 augmented)
- ChartQA val_human: 960 (Qwen)
- Two synthetic pilots: 2,400 questions

Activations

- Forward hooks inside the *same* `generate()` call that produced the labelled answer
- Every layer, 4 pooled positions: image-token mean/max, last prompt token, prompt mean

Probes

- L_2 logistic regression on one vector, scored by AUROC
- Splits by *chart*, never by question
- Figure-level bootstrap CIs
- Always next to a baseline that sees no activations

Naturalistic arm: ChartQA

Model	Overall	Human	Augmented
Qwen2.5-VL-7B	87.3	79.8	94.8
LLaVA-NeXT-7B	52.9	44.6	61.1

Relaxed accuracy (%), numeric within 5% of gold.
Qwen is wrong on **317** questions, LLaVA on **1,178**.

What Qwen's 317 errors actually are

Label	<i>n</i>
NOT_AN_ERROR	112
COMPUTATION	93
STRUCTURAL	71
AMBIGUOUS	40
FABRICATION	1

Labelled by an LLM annotator (Claude Opus 5.5) that saw only image, question, gold and answer, following a written rubric. Human agreement check not yet run.

- **Fabrication is essentially absent** from ChartQA: questions are written about content the chart contains
- **35% are not errors**: equivalent format, wrong gold, or a valid reading of the question
- COMPUTATION (inputs read right, arithmetic wrong) is the **largest real class**; the proposal's two-class taxonomy had no slot for it

⇒ ChartQA cannot supply fabrication examples.
This is why the synthetic arm exists.

Label noise in ChartQA itself

	Human	Augmented
Questions	1250	1250
Scored wrong	252	65
Real errors	140	25
Gold answer wrong	16	18

- 34 of 317 gold answers are wrong; on augmented questions that is 28% of the “errors”
- 16 answers are the right number on the other scale (0.43 for a gold of 43): the metric scores them wrong
- Counting those as correct: 87.3% → 88.0%; counting all NOT_AN_ERROR: 91.8%

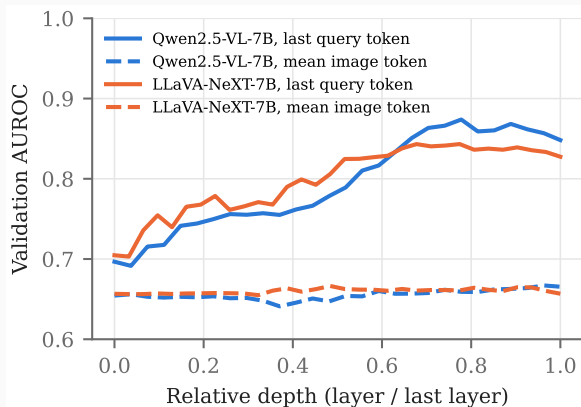
Every probe trained on relaxed-accuracy labels inherits this noise.

E1: predicting whether the answer will be wrong

Test AUROC [95% CI]	Probe	Surface baseline
Qwen, all errors	0.877 [.83, .92]	0.715
Qwen, without NOT_AN_ERROR	0.932 [.90, .96]	0.786
LLaVA, all errors	0.798 [.76, .84]	0.608
Qwen, structural vs. correct	0.905 [.84, .97]	—

- Surface baseline: question source, length, image tokens, numeric gold
- Removing label noise *raises* the probe: part of its “error” was the labels
- Not just question style: within human-written questions only, 0.895 (Qwen) and 0.850 (LLaVA)

E1: where the signal lives



- Last prompt token, building with depth, plateau in the last third (Qwen best layer 21/28, LLaVA 24/32)
- Image tokens flat at 0.62–0.68, near the surface baseline
- Differs from HALP (Qwen best on visual features for photo hallucination): for charts, the state that has read the question matters

Synthetic arm

Making fabrication happen on purpose

Ask about a category that is *not* on the chart.

- **Pilot 1:** 300 bar charts. Same wording twice per chart: “What is the value of Mango?” (shown) vs. “... of Guava?” (missing)
- **Pilot 2:** 300 harder charts, 1,800 questions
 - Lookalike pairs (Dubai/Dublin, Eagle/Falcon): chart shows at most one
 - Three families: *value*, *compare*, *neighbor* (“Which bar is right of Guava?”)
 - Values never multiples of 5, axes up to 200, sparse ticks

Two guards against leakage

- Every contrast is *within one question template* (the template alone predicts the intended type at AUROC 1.000)
- Outcomes scored from the actual answer, never from what the question was designed to cause

How the models answered

		Qwen	LLaVA
Shown, right	value	554/600	185/600
	compare	300/300	209/300
	neighbor	294/300	255/300
Missing, value	rejected	15	0
	“0”	494	232
	fabricated	91	368
Missing, compare	fabricated	37	101
Missing, neighbor	fabricated	300	290

- **Neither model refuses:** 15/1,200 (Qwen), 0 (LLaVA)
- Qwen mostly says “0”; when it invents, it reads the lookalike bar (asked Dubai, reports Dublin)
- LLaVA snaps to gridlines (52 → 50): only 31% of shown values right
- “0” is treated as a refusal expressed as a number

Value template gives four classes: C correct, S misread, F invented number, Z “0”.

Do the models know the bar is missing? Yes.

Test AUROC	Qwen	LLaVA
value	0.999	0.977
compare	0.991	0.959
neighbor	1.000	0.996
lookalikes only	0.98–1.00	0.92–0.99
text-only baseline	0.49–0.52	0.49–0.52
image-token probe	0.500	0.500

- Both models **represent that the name is missing, then answer anyway**
- Text-only baseline at chance: every name is shown on some charts
- Image-token control is exactly 0.500 by construction (image precedes the question, attention is causal)
- Early in Qwen (layer 2), later in LLaVA (~9)

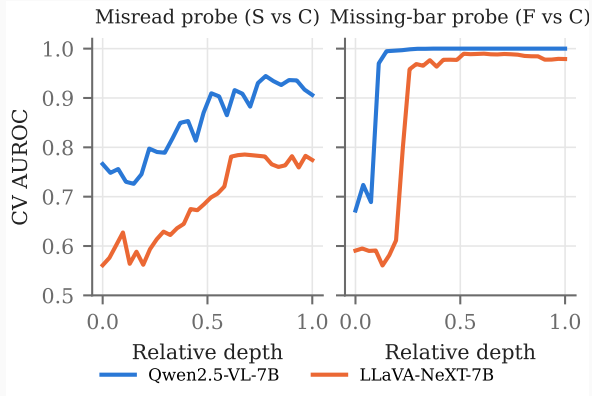
E3: cross-transfer matrix (the central result)

Probe (trained on)	Qwen2.5-VL-7B			LLaVA-NeXT-7B		
	S vs C	F vs C	F vs Z	S vs C	F vs C	F vs Z
structural (S vs C)	0.887	0.594	0.400	0.776	0.879	0.215
metadata baseline	0.814	0.557	0.543	0.512	0.522	0.537
fabrication (F vs C)	0.576	1.000	0.286	0.680	0.988	0.243
invented vs. zero (F vs Z)	0.336	0.138	0.984	0.520	0.226	0.986

- **Qwen: largely type-specific.** Each probe strong on its own task, weak on the other
- **LLaVA: misread probe is not specific.** It fires *more* on fabrications than on misreads, a general “not grounded” direction
- **Both:** the “fabrication” probe is really a missing-bar detector (scores “0” as high as invented numbers)

5-fold CV by chart, out-of-fold AUROC, synthetic value questions.

E3: where the type signals live



- Missing-bar signal saturates early (Qwen by layer 3, LLaVA ~8)
- Misread signal builds gradually, peaks in the last third
- Invented-vs-“0” only in the last layers: reads out the answer about to be produced, not an early warning

Where we stand

Hypothesis: supported for Qwen, partly for LLaVA.

- Qwen2.5-VL-7B: misread and missing-bar states are linearly distinguishable, each probe largely blind to the other type
- LLaVA-NeXT-7B: missing-bar signal is specific, but the misread probe captures a general lack of grounding; a one-number chart feature (distance to nearest gridline) predicts its misreads at 0.852, above the probe

Caveats we already know about

- On synthetic data, “fabrication” and “missing category” coincide
- Qwen has only 46 synthetic misreads; the ChartQA structural probe has 13 test positives and no surface baseline yet
- Labels are from one LLM annotator, human κ check pending

Status and next steps

Done

- Inference and full-depth caches: both models, ChartQA test, val_human (Qwen), both pilots
- 317 Qwen error labels
- E1 both models; E3 both models

Coded, waiting for the cluster

(maintenance; the activation caches live there)

- Synthetic $\rightarrow$ ChartQA transfer: does the synthetic misread probe flag the 71 real misreads?
- COMPUTATION as a third class vs. misreads
- E4 intervention matrix

Timeline

- **Oct 1–8:** human κ on 200 items; LLaVA's 1,178 error labels; residualised probes (E2)
- **Oct 1–15:** E4 repairs (crop, verify prompt, abstain) on synthetic items, both models
- **Oct 16–25:** E5 controller; risk–coverage vs. binary selective prediction and the oracle
- **Oct 26–31:** final report

Where we would like guidance

1. **Fabrication definition.** Is “asked about a missing category” an acceptable operationalisation, or should we add out-of-range quantities so fabrication is not identical to absence?
2. **Taxonomy.** Should COMPUTATION become a third type in the hypothesis, given it is the largest real error class?
3. **Annotation.** Is an LLM annotator acceptable if it passes a two-person human agreement check ($\kappa > 0.6$ on 200 items)?
4. **Label noise.** Should the 112 NOT_AN_ERROR items be re-scored as correct in every experiment, or reported both ways?
5. **Scope of E4/E5.** Is running the repairs on synthetic charts only (where bar geometry is known) enough, or do we need ChartQA repairs too?

Thank you